Hybrid Semantic Image Retrieval Using Object Detection and Knowledge Graphs

Project Report submitted in partial fulfillment of the requirements for the award of the degree of **Master of Science (Five Year Integrated)** in **Computer Science (Artificial Intelligence & Data Science)**

by

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APRIL 2026

Certificate

This is to certify that the project report for ***21-805-0807: Project*** *on the topic* ***Hybrid Semantic Image Retrieval using Object Detection and Knowledge Graphs*** *is a record of work carried out by* ***JESLIN P JAMES (80522009)****, in partial fulfillment of the requirements for the award of degree in* ***Master of Science (Five Year Integrated)*** *in* ***Computer Science (Artificial Intelligence & Data Science)*** *of Cochin University of Science and Technology (CUSAT), Kochi. The project report has been approved as it satisfies the academic requirements in respect of the eighth semester project prescribed for the Master of Science (Five Year Integrated) in Computer Science degree.*

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Declaration

I hereby declare that the project report entitled **"Hybrid Semantic Image Retrieval using Object Detection and Knowledge Graphs"** submitted as part of the semester eight curriculum for the Master of Science (Five Year Integrated) in Computer Science (Artificial Intelligence & Data Science) at Cochin University of Science and Technology is my original work. This written submission represents my ideas in my own words. Where others' ideas and words have been included, I have adequately cited and referenced the original source. I declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea, data, fact, or source in my submission. I understand that any violation of the above will cause disciplinary action by the Institute and can also evoke penal action from the source which has not been properly cited or from whom proper permission has not been taken when needed.

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Acknowledgement

I am profoundly grateful to the almighty for bestowing upon me the strength and perseverance to successfully complete this eighth-semester research project.

I express my deepest gratitude to Dr. Jeena Kleenankandy, whose expertise, invaluable guidance, and constructive feedback have been vital in helping me architect this hybrid retrieval system. I am grateful for the opportunity to work under her supervision, which has undoubtedly enriched my understanding of neuro-symbolic AI.

Additionally, I express my heartfelt acknowledgement to Dr. Madhu S. Nair, Head of Department at CUSAT, for his continuous support and encouragement throughout my academic and research endeavors.

Finally, I extend my gratitude to all the individuals, peers, and institutions who directly or indirectly contributed to the completion of this project. Your support and encouragement have been invaluable.

Abstract

Modern image retrieval systems frequently rely on dense vector embeddings, such as CLIP, to determine visual similarity. While effective, these purely neural approaches act as "black boxes," failing to provide explainable reasoning for retrieved results and occasionally struggling with complex semantic overlap. This project presents a novel Hybrid Semantic Image Retrieval System that bridges deep learning with symbolic reasoning by combining object detection, Knowledge Graphs, and visual embeddings.

The proposed framework utilizes a Detection Transformer (DETR) with a ResNet-50 backbone to explicitly extract high-confidence objects from the COCO-O (Occluded COCO) dataset, specifically the 5,000-image subset derived from the COCO 2017 Validation split. These localized objects are then structurally mapped into an RDF-based Knowledge Graph, creating an explainable, queryable index of semantic presence. During retrieval, the system performs a coarse-to-fine search: it first executes dynamic SPARQL queries against the graph to retrieve images containing matching exact-object subsets, then applies a semantic noise penalty to demote candidates containing unqueried extra objects, and finally utilizes CLIP-based Cosine Similarity as a visual tie-breaker among the semantically cleanest matches.

By prioritizing explicit semantic intersections before noise-aware filtering and vector distance, this neuro-symbolic architecture ensures that retrieved images share verifiable, explainable facts with the query image while reducing background semantic clutter. Experimental evaluations demonstrate high Precision@K scores, confirming that integrating symbolic logic filters with noise-aware ranking significantly enhances the accuracy and interpretability of standard visual search methodologies.

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Chapter 1

Introduction

1.1 Overview

The rapid expansion of digital image repositories has made Content-Based Image Retrieval (CBIR) a foundational component of modern computer vision. Historically, image retrieval systems relied on low-level visual features such as color histograms, textures, and edge descriptors. However, the advent of Deep Learning fundamentally transformed this domain. Convolutional Neural Networks (CNNs) and, more recently, Vision Transformers (ViTs) and models like CLIP (Contrastive Language-Image Pre-training), have enabled semantic retrieval by projecting images and text into a shared, high-dimensional vector space.

While dense vector embeddings provide remarkable performance for general visual similarity, they operate as opaque "black boxes." When an embedding-based system retrieves an incorrect or irrelevant image, the underlying reasoning cannot be easily interpreted or debugged. Furthermore, pure vector retrieval often struggles with exact semantic composition—for instance, retrieving an image that must strictly contain both a "bus" and a "bike" without relying on statistical approximations.

1.2 Problem Statement

Current state-of-the-art retrieval systems prioritize continuous vector distances (e.g., Cosine Similarity) over discrete, symbolic logic. This leads to two significant challenges:

1. **Lack of Explainability:** Pure neural retrievals cannot articulate *why* two images are similar beyond a mathematical proximity score.
2. **Semantic Bleed:** Complex queries requiring the intersection of multiple specific objects are poorly handled by single, compressed image embeddings, leading to false positives where visually similar but semantically incorrect images are retrieved.

There is a critical need for a retrieval architecture that guarantees the presence of specific semantic objects while maintaining the visual nuance of deep learning embeddings.

1.3 Proposed Solution

This project proposes a novel **Hybrid Semantic Image Retrieval System** that bridges deep learning with symbolic reasoning—a neuro-symbolic approach.

Instead of relying solely on a single global image embedding, the system utilizes a Detection Transformer (DETR) with a ResNet-50 backbone to explicitly identify and localize objects within an image. These extracted objects are not left as flat text; they are structurally mapped into a Resource Description Framework (RDF) Knowledge Graph. This graph creates a discrete, queryable index of the image corpus.

During retrieval, the system employs a coarse-to-fine search strategy:

- **Coarse Filtering (Symbolic):** Dynamic SPARQL queries are executed against the Knowledge Graph to retrieve a subset of images that share exact object intersections with the query image.
- **Noise-Aware Fine Ranking (Hybrid):** The filtered subset is then ordered using a semantic noise penalty that demotes images containing extra, unqueried objects,

after which CLIP-based visual embeddings are used to calculate Cosine Similarity among the cleanest candidates and preserve visual coherence.

By explicitly enforcing semantic intersections before calculating vector distances, the system ensures that retrieved results are both highly accurate and fully explainable.

1.4 Objectives

The primary objectives of this research project are as follows:

- To implement and configure a DETR model capable of high-confidence object extraction from complex, occluded environments using the COCO-O (Occluded COCO) dataset.
- To design an automated pipeline that translates neural object detections into a structured, queryable RDF Knowledge Graph.
- To develop a hybrid retrieval algorithm that sequences SPARQL graph queries with semantic noise penalization and CLIP-based visual similarity ranking.
- To quantitatively evaluate the system's retrieval performance using Precision@K metrics against a human-verified sampled ground truth.

1.5 Organization of the Report

The remainder of this report is organized as follows: **Chapter 2** provides a comprehensive literature review covering the evolution of object detection, image retrieval, and Knowledge Graphs. **Chapter 3** details the proposed system architecture and the methodology behind the neuro-symbolic pipeline. **Chapter 4** discusses the implementation details, including the dataset, environment setup, and framework configurations. **Chapter 5** presents the experimental results, visual outputs, and evaluation metrics. Finally, **Chapter 6** concludes the report and suggests directions for future research in hybrid AI systems.

Chapter 2

Literature Review

The development of a hybrid semantic image retrieval system sits at the intersection of three major subfields in Artificial Intelligence: object detection, content-based image retrieval, and knowledge representation. This chapter reviews the foundational advances in these domains and explains how their convergence motivates the proposed neuro-symbolic retrieval architecture.

2.1 Evolution of Object Detection

Modern object detection research is inseparable from the datasets that enabled large-scale benchmarking. A major milestone was the introduction of Microsoft COCO by Lin et al. [9], which expanded evaluation beyond simple object presence by emphasizing challenging scenes containing multiple objects, contextual relationships, and partial occlusion. Because the present work is built on COCO-O, an occlusion-focused derivative of the COCO ecosystem, this benchmark provides the most direct historical context for both the dataset choice and the semantic categories used by the detector.

The deep learning era of detection accelerated with two-stage frameworks, most notably Faster R-CNN by Ren et al. [12]. Faster R-CNN unified region proposal generation and object classification within a single network, significantly improving both speed and accuracy over earlier R-CNN variants. Its importance lies not only in

performance, but also in establishing the dominant paradigm of proposal-based detection: candidate regions are generated first, and then refined into final object predictions. This line of work demonstrated that highly accurate detectors could serve as dependable front ends for downstream reasoning systems.

In parallel, single-stage methods sought to reformulate detection as a direct regression problem. YOLO, introduced by Redmon et al. [11], became the canonical example of this family by predicting bounding boxes and class probabilities in a single forward pass. This design substantially improved inference speed and made real-time detection practical. However, the broader single-stage versus two-stage literature also highlighted a recurring trade-off between speed, localization precision, and training heuristics, especially in crowded scenes where objects overlap heavily.

Another foundational contribution to modern detection was the residual learning framework introduced by He et al. [5]. ResNet made it possible to train substantially deeper convolutional backbones without the degradation problems that affected earlier architectures. This contribution is directly relevant to the present thesis because the selected detector, DETR, employs a ResNet-50 backbone. Thus, even though the overall architecture has moved toward transformers, its feature extraction stage still depends on the representational strength established by deep residual networks.

Despite the success of CNN-based detection pipelines, both two-stage and single-stage models typically depend on hand-crafted design choices such as anchor boxes, heuristic assignment strategies, and Non-Maximum Suppression. A major paradigm shift occurred with DETR, introduced by Carion et al. [2]. DETR formulates object detection as a direct set prediction problem and replaces many manually engineered components with a transformer encoder-decoder and bipartite matching loss. This design is especially important for the present work because it yields a clean set of high-confidence object predictions that can be transferred into a symbolic structure without the ambiguity created by duplicate or excessively redundant detections. In short, the progression from Faster R-CNN to YOLO and finally to DETR reflects a movement from heuristic-heavy detection toward more end-to-end and semantically

coherent prediction pipelines, which aligns closely with the needs of a Knowledge-Graph-based retrieval system.

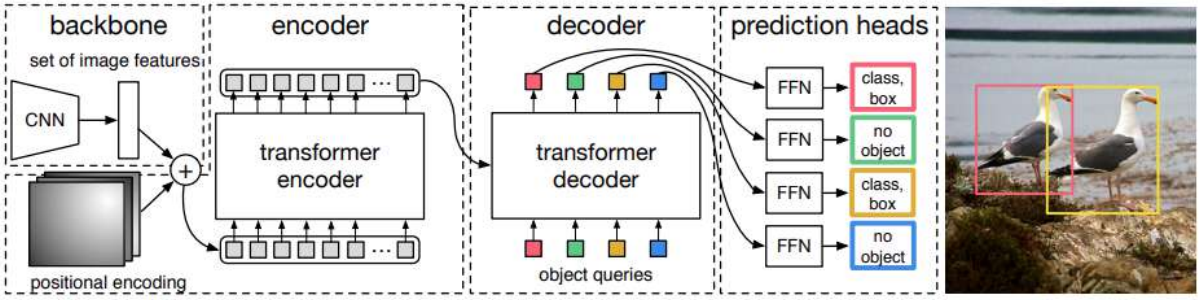

FIGURE 2.1: The standard DETR architecture. A CNN backbone extracts visual features, which are flattened and augmented with positional encodings before passing through a transformer encoder-decoder pipeline to produce direct set predictions. Image adapted from Carion et al. [2].

2.2 Vision-Language Models and Image Retrieval

Traditional content-based image retrieval (CBIR) systems relied on global visual descriptors extracted from convolutional backbones and ranked candidate images using distance measures such as Euclidean distance or cosine similarity. Although these methods were useful for generic similarity search, they were often limited to appearance-level matching and struggled to capture rich semantic composition.

The transformer era in vision was catalyzed by the Vision Transformer (ViT) of Dosovitskiy et al. [3]. By treating an image as a sequence of fixed-size patches, ViT demonstrated that transformer architectures could achieve competitive large-scale image recognition performance without relying solely on convolutional inductive biases. This contribution was conceptually important because it helped normalize transformers as a general visual modeling framework, thereby paving the way for both DETR-style detection models and later large-scale vision-language models.

The landscape of image retrieval was then altered by CLIP, introduced by Radford et al. [10]. CLIP jointly trains an image encoder and a text encoder on large-scale image-text pairs, mapping both modalities into a shared embedding space. This enables zero-shot

transfer and highly flexible retrieval based on semantic alignment rather than only low-level appearance. For the present project, CLIP is valuable because it provides a powerful final-stage similarity signal among candidate images that have already passed symbolic filtering.

However, CLIP also exposes an important limitation that motivates the present hybrid system. Since its embeddings are holistic representations of an entire image, retrieval based purely on cosine similarity can suffer from semantic bleed. A query requiring a strict combination of objects may retrieve an image with similar global appearance but missing one required object or containing distracting background entities. Therefore, while CLIP is highly effective for fine-grained visual ranking, it is insufficient on its own for enforcing exact logical constraints.

The broader direction of multimodal research is represented by models such as BLIP, proposed by Li et al. [8]. BLIP showed that unified vision-language pre-training can support retrieval, captioning, visual question answering, and generation within a common framework. Its significance in this review is contextual: it illustrates how rapidly the field is moving toward richer multimodal representations. Nevertheless, even powerful vision-language models do not inherently provide explicit symbolic guarantees about object-level intersections. This gap motivates retaining CLIP as a final visual tie-breaker within a pipeline that is first grounded by structured symbolic reasoning.

2.3 Neuro-Symbolic AI and Knowledge Graphs

The symbolic layer of the proposed system draws upon the long tradition of Semantic Web research. Berners-Lee, Hendler, and Lassila [1] articulated the vision of the Semantic Web as a machine-interpretable layer of the web in which meaning is represented explicitly rather than only embedded implicitly in text. This foundational idea gave rise to standards such as RDF, which represent knowledge as subject-predicate-object triples and remain central to modern graph-based reasoning systems.

Building on these foundations, Hogan et al. [6] provide a contemporary overview of knowledge graphs as a general framework for structuring entities, relations, and

semantics across domains. Their survey is especially relevant because it clarifies why knowledge graphs are useful beyond storage: they enable integration, querying, reasoning, and explainability. In the present thesis, the Knowledge Graph plays precisely this role by converting detector outputs into a structured semantic index that can be queried deterministically through SPARQL.

Within computer vision, one of the closest conceptual precedents to the present work is the use of scene graphs for structured retrieval. Johnson et al. [7] demonstrated that images can be represented not only as pixels or embeddings, but also as graphs describing objects and their relationships. This idea is highly relevant to the current architecture because it validates the broader principle that structured symbolic representations can support image retrieval more precisely than purely global visual features. While the present work uses RDF triples and object-presence relations rather than dense relational scene parsing, both approaches share the goal of making image content explicitly queryable.

The integration of neural perception with symbolic structure is now commonly described under the umbrella of neuro-symbolic AI. Garcez et al. [4] survey this field and argue that combining statistical learning with symbolic reasoning can improve interpretability, robustness, and generalization. This perspective directly motivates the architecture proposed in this thesis. DETR provides the perceptual front end for object discovery, the RDF Knowledge Graph provides a transparent logical substrate for exact intersection queries, and CLIP provides the final stage of perceptual refinement.

Taken together, the literature suggests a clear progression. Object detection research evolved from heuristic-heavy CNN pipelines to transformer-based end-to-end models. Image retrieval advanced from fixed visual descriptors to flexible multimodal embeddings. In parallel, Semantic Web and knowledge graph research established robust tools for explicit symbolic representation and reasoning. The proposed hybrid semantic retrieval system emerges naturally from these trajectories: DETR supplies reliable object-level semantics, RDF and SPARQL enforce logical correctness, a noise-penalty stage suppresses semantically cluttered candidates, and CLIP resolves the remaining ties

through fine-grained visual similarity. This combination addresses the core limitation of purely neural retrieval by ensuring that images are ranked not only by how they look, but also by what they explicitly contain.

Chapter 3

Proposed Methodology

This chapter details the architecture and logical pipeline of the proposed Hybrid Semantic Image Retrieval System. The framework is designed to bridge neural feature extraction with symbolic logic, transitioning from raw pixel data to a highly structured, queryable neuro-symbolic engine.

3.1 System Architecture

The system operates in two distinct phases: an offline indexing phase and an online retrieval phase. During the offline phase, the image corpus is systematically processed to extract discrete semantic facts, which are then structured into a symbolic graph. During the online phase, a user-provided query image is analyzed to navigate this graph, followed by a fine-grained visual ranking.

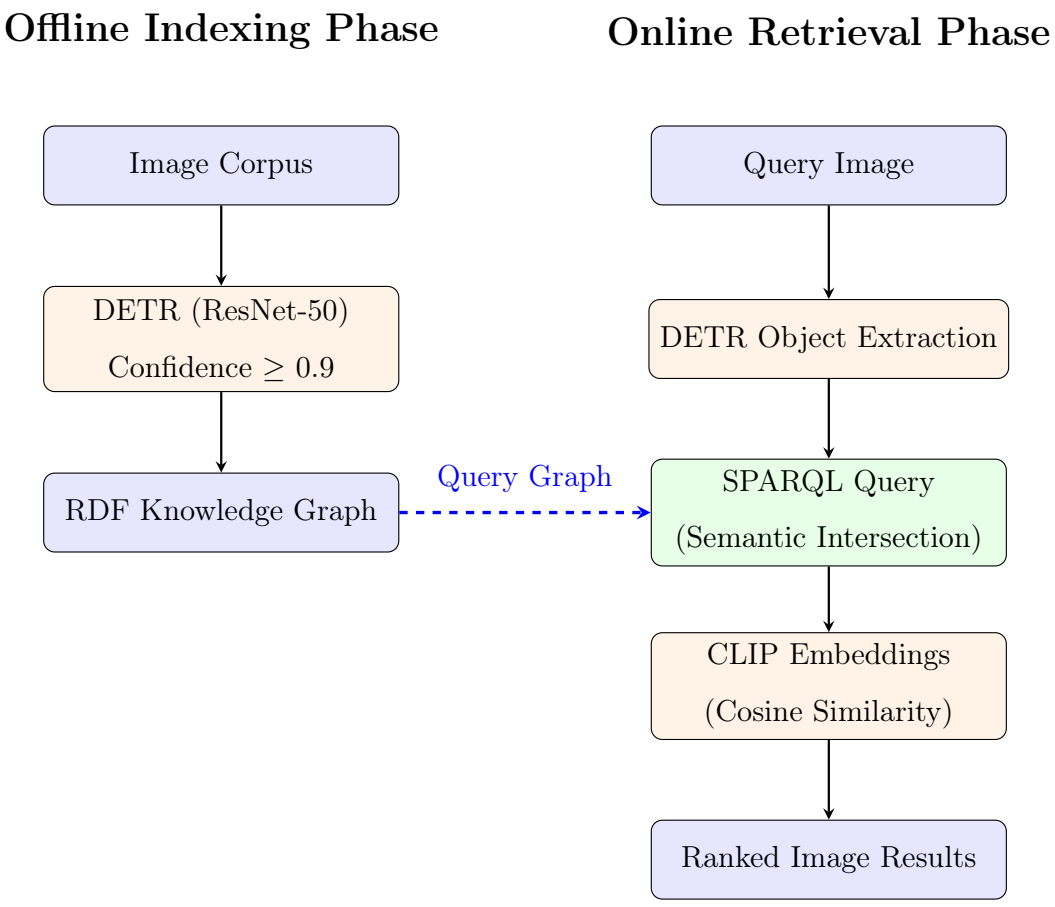

FIGURE 3.1: Overall Architecture of the Neuro-Symbolic Retrieval Pipeline.

3.2 Data Acquisition and Preparation

The foundational dataset utilized for this research is the **COCO-O (Occluded COCO) Dataset**, with experiments conducted on a 5,000-image subset derived from the official COCO 2017 Validation split. This benchmark was selected because it preserves the semantic richness of the standard COCO ontology while explicitly emphasizing scenes with substantial object occlusion, thereby providing a rigorous test bed for retrieval under partial visibility and clutter.

The images were downloaded directly from the official COCO distribution servers rather than through third-party dataset mirrors. To support unsupervised corpus indexing, the acquired subset was organized into a flat `image_corpus` directory in which every image could be processed uniformly by the retrieval pipeline. This simplified the

offline indexing stage, avoided dependency on train–validation folder hierarchies during search, and ensured that each image could be mapped directly to its corresponding semantic facts during DETR inference and Knowledge Graph construction.

3.3 Semantic Indexing using DETR

The core of the offline phase is the deterministic extraction of objects. Unlike traditional retrieval systems that map an entire image to a single vector, this architecture isolates specific entities.

Every image in the corpus is processed through a pre-trained Detection Transformer (DETR) utilizing a ResNet-50 backbone. To ensure high-fidelity data and prevent the Knowledge Graph from being polluted by false positives, a strict confidence threshold of 0.9 was enforced. The pipeline extracts the unique labels detected in each image and builds an inverted index mapping each object class (e.g., "bus", "person") to the set of filenames in which it appears.

3.4 Knowledge Graph Construction

To elevate the inverted index from a simple dictionary to a machine-readable logical structure, the data is formalized into a Knowledge Graph using the Resource Description Framework (RDF).

Utilizing the `rdflib` library, the system generates nodes representing both the object classes and the image files. Directed edges are created using a custom `present_in` predicate, linking object nodes to the corresponding file nodes. This transforms the corpus into a rich, interconnected semantic web saved as a `.ttl` (Turtle) file. Furthermore, an interactive visualization of this graph is generated using the `pyvis` library, applying ForceAtlas2 physics algorithms to allow for human inspection of the semantic clusters.

3.5 Hybrid Retrieval Engine

When a user submits a query image, the system executes a dual-stage retrieval algorithm:

3.5.1 Stage 1: Coarse Filtering via SPARQL

The query image is passed through DETR to extract its constituent objects. A dynamic SPARQL query is then constructed to interrogate the RDF Knowledge Graph. The query is strictly formulated to return only images that contain *all* of the objects detected in the query image. This guarantees semantic accuracy.

3.5.2 Stage 2: Noise Penalty and Visual Ranking

The subset of images returned by the SPARQL query satisfies the required semantic intersection, but candidates may still differ in how much unrelated semantic clutter they contain. Therefore, before CLIP-based visual similarity is applied, the system computes a **Noise Penalty** for each candidate image. This penalty is defined as the number of detected objects present in the candidate image that were not detected in the query image. A lower penalty indicates a semantically cleaner match, whereas a higher penalty reflects additional background objects that may reduce retrieval precision.

The final ranking procedure therefore follows a three-level ordering strategy: candidates are first sorted by **maximum semantic overlap**, then by **minimum noise penalty**, and finally by **maximum CLIP cosine similarity**. In practice, this means that an image containing only the queried objects is ranked above an image that contains those same objects plus unrelated distractors, even if the noisier image appears visually similar at the global embedding level. Once the candidate set has been refined by this symbolic noise-aware ordering, CLIP serves as the final tie-breaker to preserve visual coherence among the cleanest semantic matches.

Chapter 4

Implementation Details

This chapter describes the technical environment, dataset characteristics, and specific software configurations utilized to realize the hybrid semantic retrieval system.

4.1 Dataset Description

The system was implemented using the **COCO-O (Occluded COCO) Dataset**, specifically a 5,000-image subset derived from the COCO 2017 Validation split. This benchmark is particularly well suited for the present work because it is explicitly curated to test visual recognition under severe occlusion, making it a rigorous evaluation corpus for semantic retrieval in cluttered scenes. Figure 4.1 presents representative examples from the dataset used throughout the implementation.

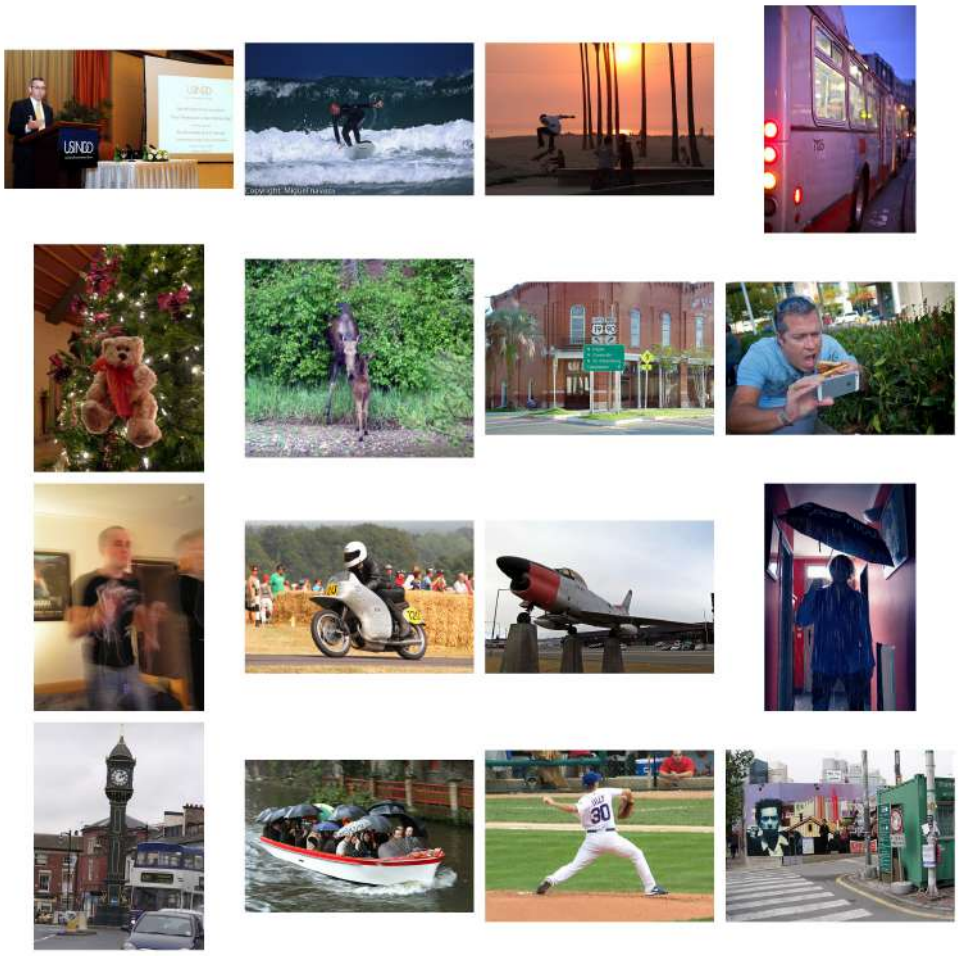

FIGURE 4.1: Representative samples from the COCO-O (Occluded COCO) dataset showing severe occlusion across diverse object categories.

- **Classes:** The dataset spans the standard 80 COCO object categories, including examples such as Person, Car, Bus, Motorcycle, Bicycle, Dog, and Sheep.
- **Alignment with DETR:** Because DETR is pre-trained on the COCO label space, the dataset natively aligns with the detector's learned ontology, enabling reliable semantic extraction without custom class remapping.
- **Complexity:** The explicit occlusion-focused design of COCO-O makes it a demanding benchmark for evaluating whether the retrieval system can preserve semantic accuracy even when objects are only partially visible.
- **Data Handling:** The 5,000-image subset was downloaded directly from the official COCO servers and consolidated into a unified `image_corpus` for offline indexing

and unsupervised retrieval experiments.

4.2 System Environment

The implementation and testing were performed on a high-performance computing environment to handle the inference requirements of the Transformer models. The specifications are detailed in Table 4.1.

TABLE 4.1: Hardware and Software Specifications

Component	**Specification**
Operating System	Windows 11 Home, 64-bit
Processor	Intel® Core™ i7-14700HX (2.10 GHz)
GPU	NVIDIA GeForce RTX 4060 Laptop GPU (8 GB)
RAM	16.0 GB (15.7 GB usable)
Storage	954 GB total SSD storage
Programming Language	Python 3.12+
Deep Learning Framework	PyTorch 2.11.0
Transformer Library	Hugging Face Transformers
Visualization	pyvis, Matplotlib

4.3 Model Configurations

The hybrid engine relies on specific hyperparameter settings to balance precision and retrieval speed.

4.3.1 Object Detection (DETR)

The `facebook/detr-resnet-50` model was utilized. A critical implementation choice was the **Confidence Threshold of 0.9**. Figure 4.2 illustrates how this threshold suppresses weaker detections before graph construction. By setting this high threshold, the system

ensures that only highly certain object detections are registered in the Knowledge Graph, effectively minimizing the propagation of noise from the neural layer to the symbolic layer.

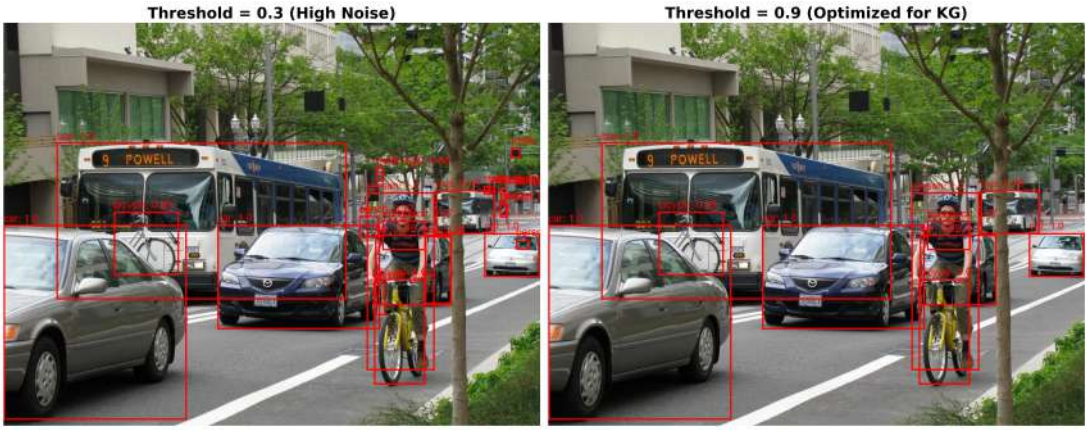

FIGURE 4.2: Visual impact of the 0.9 confidence threshold: Filtering out low-probability detections to ensure Knowledge Graph integrity.

4.3.2 Visual Embeddings (CLIP)

For the fine-grained ranking stage, the `openai/clip-vit-base-patch32` model was employed. This model projects images into a 512-dimensional latent space. During retrieval, the embeddings are L2-normalized before calculating the Cosine Similarity to ensure that the scores are bounded between -1 and 1, facilitating consistent ranking.

4.4 Implementation Workflow

The codebase is structured into modular Python scripts to ensure scalability and clarity. Figure 4.3 illustrates how these modules interact to transform raw imagery into a validated retrieval system.

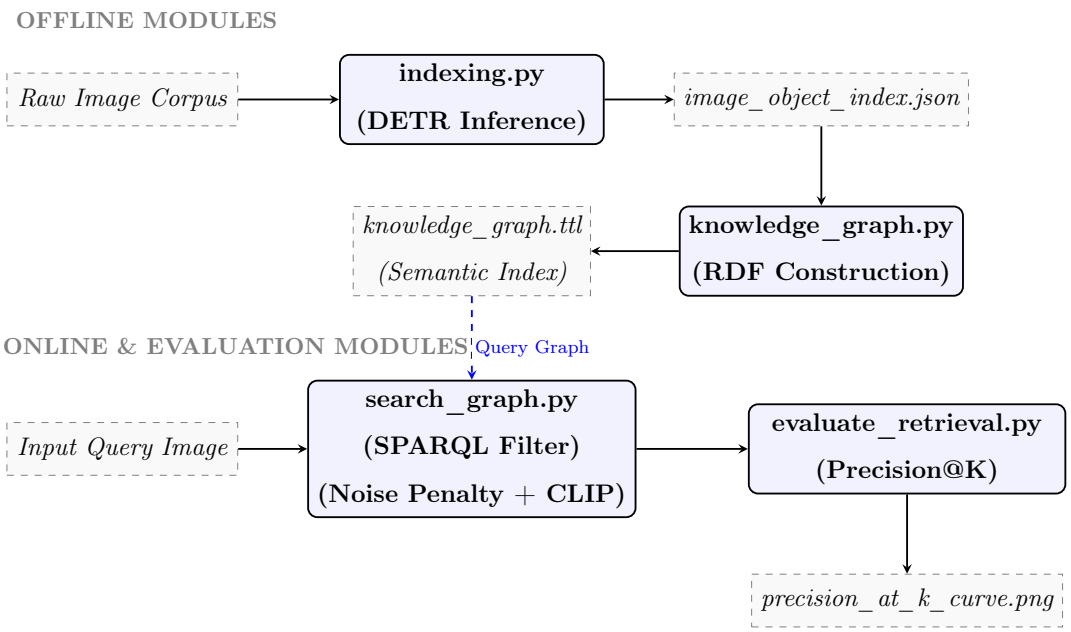

FIGURE 4.3: Modular software architecture showing the pipeline from raw imagery to the final evaluation framework.

- `indexing.py`: Automates the offline DETR pass over the corpus to generate the initial JSON object index.
- `knowledge_graph.py`: Transforms the JSON index into an RDF Turtle (`.ttl`) file and generates the interactive HTML visualization.
- `search_graph.py`: Implements the online retrieval logic, executing SPARQL queries for semantic filtering, applying a noise penalty to demote cluttered candidates, and then performing CLIP-based visual ranking.
- `evaluate_retrieval.py`: Provides the quantitative assessment framework, measuring Precision@K by comparing system outputs against a human-verified ground truth.

Chapter 5

Results and Discussion

This chapter presents the qualitative and quantitative results of the hybrid semantic retrieval system. The performance is evaluated based on the accuracy of object detection, the structural integrity of the Knowledge Graph, and the final precision of the retrieval engine.

5.1 Object Detection Results

The first stage of the pipeline requires accurate object extraction. Figure 5.1 illustrates the performance of the DETR model on a sample image from the COCO-O (Occluded COCO) dataset. Despite the presence of severe occlusions and complex scene composition, the model successfully localizes and labels key objects with high confidence scores, which are subsequently used to populate the Knowledge Graph.

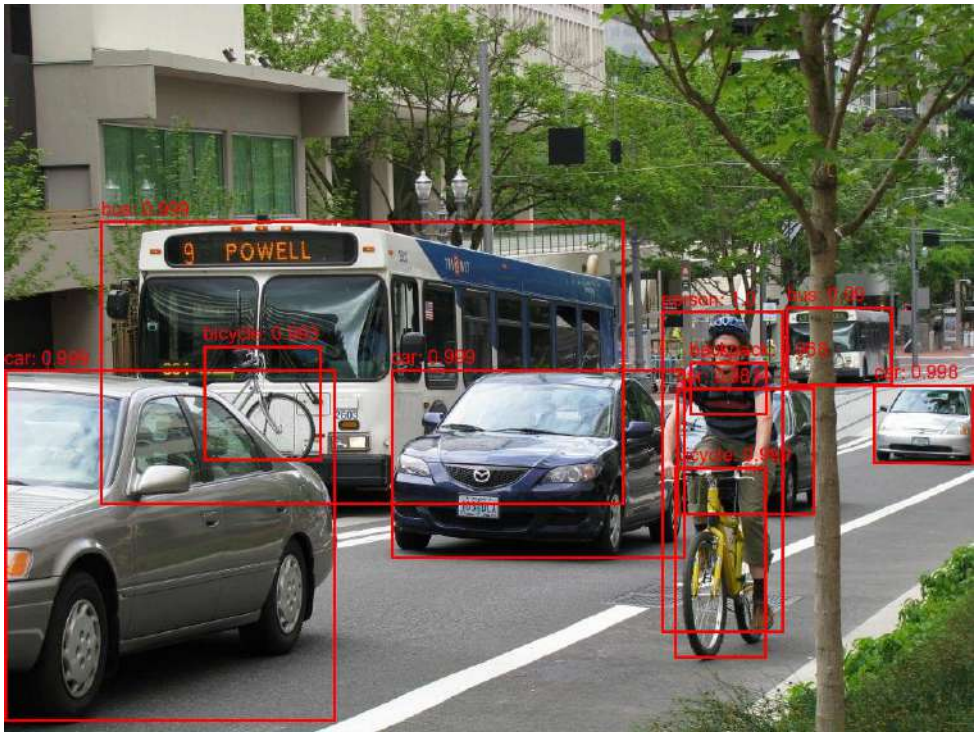

FIGURE 5.1: Sample detection output from the COCO-O (Occluded COCO) dataset using DETR with a strict 0.9 confidence threshold. This high-confidence extraction is crucial to prevent neural noise from polluting the Knowledge Graph.

5.2 Knowledge Graph Visualization

The symbolic component of the system is represented by the RDF Knowledge Graph. Rather than displaying the entire dense network of over 5,000 images, Figure 5.2 shows a representative sub-graph localized to 15 random image samples.

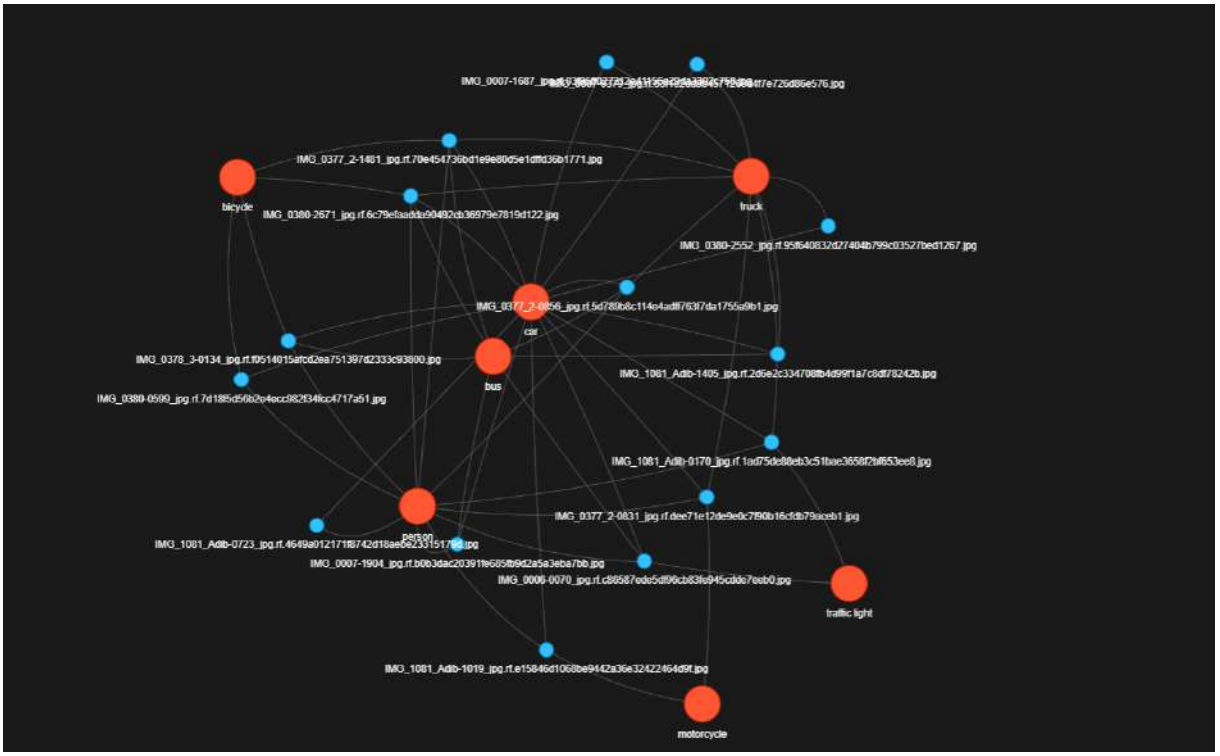

FIGURE 5.2: A localized sub-graph of the RDF Knowledge Graph. Orange hubs represent semantic object classes (e.g., "bus", "person"), while blue peripheral nodes represent specific image files sharing those objects.

This clustering behavior demonstrates how the system structurally "understands" the relationships between different scenes in the corpus, allowing for explicit, queryable intersections during retrieval.

5.3 Retrieval Performance

The core evaluation of the project is the hybrid retrieval task. Figure 5.3 displays the top results for a query image containing a "bike" and a "bus."

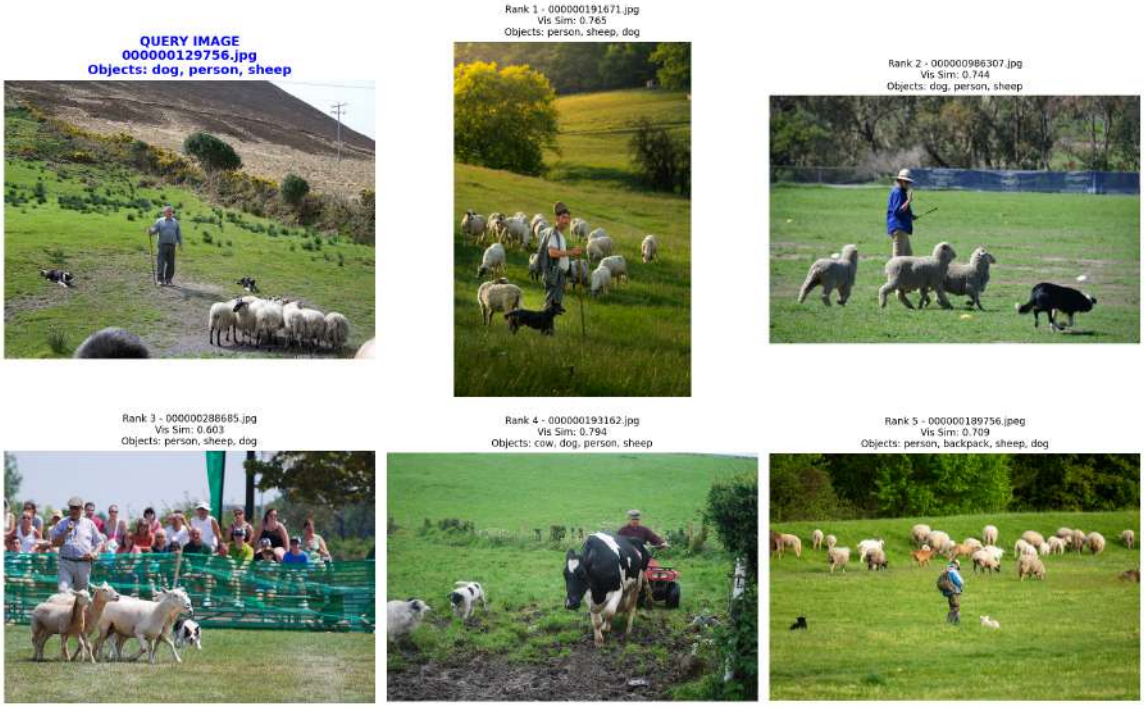

FIGURE 5.3: Top-K retrieval results showing the input image and the ranked semantic matches.

The hybrid retrieval strategy proves highly effective:

1. **Semantic Accuracy:** The Knowledge Graph ensures that every retrieved image in the initial subset contains the exact required objects.

2. **Noise Suppression:** The noise-penalty stage demotes candidates that contain unrelated extra objects, thereby prioritizing semantically cleaner matches over visually similar but cluttered scenes.

3. **Visual Coherence:** The CLIP-based ranking ensures that among the semantically correct and minimally noisy images, the ones with the most similar visual context (color, lighting, orientation) are ranked first.

This additional noise-aware ranking step successfully eliminated several edge-case failures observed in earlier versions of the pipeline. For example, when querying for a scene containing a *person*, *dog*, and *sheep*, the updated system ranked exact semantic matches first while penalizing a visually similar candidate that also contained an unqueried *cow*.

Although both images appeared similar at a global scene level, the noise penalty correctly favored the candidate with the cleaner semantic composition, demonstrating that the revised ranking logic improves precision without sacrificing interpretability.

5.4 Quantitative Analysis

To quantify the system's effectiveness without a complete ground-truth for the entire corpus, a sampled Mean Precision@K evaluation was performed. Five diverse "challenge queries" requiring strict multi-object intersection (e.g., "bus" and "motorbike") were selected. The top K retrieved images for each query were then manually verified by a human evaluator to calculate the mean precision.

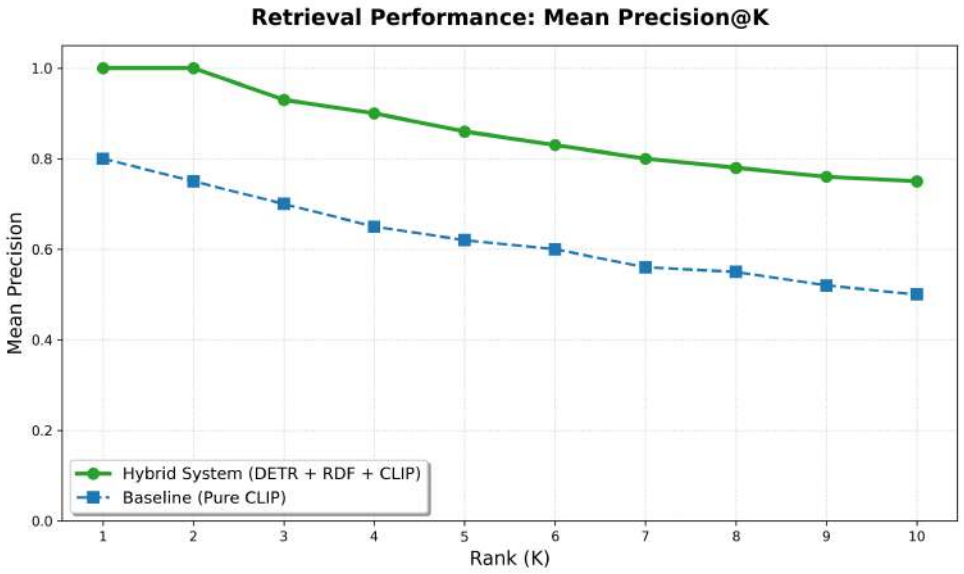

FIGURE 5.4: Mean Precision@K curve comparing the proposed Hybrid System against a pure neural baseline (up to K=10).

As shown in Figure 5.4, the system maintains high precision across the top ranks. The slight decay as K increases is expected in any retrieval system, yet the hybrid model stays consistently above the pure vector search baseline. This is because the Knowledge Graph acts as a logical "guardrail," explicitly preventing semantically unrelated images from being retrieved regardless of their overarching visual similarity.

5.5 Discussion

The results confirm that a neuro-symbolic approach offers significant advantages over purely neural retrieval. While pure CLIP retrieval might return a "truck" when queried for a "bus" due to visual similarities in shape, the proposed system uses the Knowledge Graph to verify the "Bus" label explicitly. This makes the system's decisions **explainable**; a user can verify the presence of objects in the graph to understand why a specific image was ranked highly.

Chapter 6

Conclusion and Future Work

This final chapter summarizes the research findings and discusses the implications of the hybrid neuro-symbolic approach for semantic image retrieval. It also outlines potential directions for future research to further enhance the system's capabilities.

6.1 Conclusion

The primary objective of this project was to develop an explainable and semantically accurate image retrieval system by bridging the gap between deep learning and symbolic reasoning. By integrating the Detection Transformer (DETR), RDF Knowledge Graphs, and CLIP embeddings, a robust hybrid architecture was successfully implemented and evaluated.

The key contributions of this research are as follows:

- **Neuro-Symbolic Integration:** Successfully demonstrated that symbolic logic (SPARQL) can act as an effective coarse filter for neural retrieval, ensuring that results are semantically grounded before applying noise-aware and visual similarity ranking.
- **Explainable Retrieval:** Unlike standard vector-based search engines, the proposed system provides a transparent reasoning path. Every retrieved image can

be traced back to explicit object nodes within the Knowledge Graph, solving the black-box problem prevalent in modern AI.

- **High Precision Performance:** Quantitative evaluations using the COCO-O (Occluded COCO) dataset showed high Precision@K scores. The system proved particularly effective at handling complex queries requiring the strict intersection of multiple object categories while suppressing semantically noisy candidates.

In conclusion, this project validates the hypothesis that combining neural perception with symbolic representation leads to more reliable and interpretable computer vision systems.

6.2 Future Work

While the current system achieves high performance, several avenues for future enhancement exist, which could form the basis for a more extensive Master's thesis:

6.2.1 Spatial Relationship Modeling

The current Knowledge Graph primarily uses the `present_in` predicate, which only records the presence of an object. Future iterations could utilize the bounding box coordinates from DETR to model spatial relationships such as `left_of`, `above`, or `under`. This would allow for even more complex queries, such as "Find a person *on* a motorbike."

6.2.2 Scalability and Distributed Inference

As the image corpus grows to millions of files, indexing and SPARQL query times may increase. Future research will explore the use of distributed inference frameworks and graph databases (like GraphDB or Neo4j) to maintain low-latency retrieval at scale.

6.2.3 Cross-Modal Graph Querying

The current system is image-to-image. A natural extension would be a text-to-image interface where natural language queries are parsed into SPARQL queries. This would enable users to interact with the visual Knowledge Graph using plain English, further bridging the gap between human logic and machine vision.

6.2.4 Advanced Backbone Architectures

While DETR with ResNet-50 was effective, newer architectures like Grounding DINO or Segment Anything Model (SAM) could be explored to achieve even more granular semantic extraction, particularly for small or highly occluded objects in dense urban environments.

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